PROJECT MODE

DISCOVERY DOSSIER

TO: Dr. David Kipping (Cool Worlds Lab) / NASA Exoplanet Science Institute

FROM: Saurav Kumar, Principal Investigator

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SUBJECT: Topological Detection of Non-Periodic Anomaly in Kepler-186 (D■ ≈ 1.95)

1. EXECUTIVE SUMMARY

An autonomous AI-driven pipeline utilizing **Takens' Time-Delay Embedding** has identified a significant topological anomaly in the light curve of **Kepler-186**.

While standard Fourier-based pipelines filter for strict periodicity, our attractor reconstruction reveals a low-dimensional **Strange Attractor** with a Correlation Dimension (D■) of **1.9476**. This metric distinguishes the system from both simple limit cycles (D■ ≈ 1.0) and stochastic noise (D■ > 2.5), suggesting *active multi-body dynamics* or *Transit Timing Variations (TTVs)* previously obscured by linear filtering.

Key Finding: Kepler-186 is the ONLY target in our sample that exhibits strange attractor behavior. This 5-planet system, which includes Earth-size Kepler-186f in the habitable zone, shows hidden orbital dynamics invisible to standard frequency analysis.

2. VISUAL PROOF: PHASE SPACE RECONSTRUCTION

File: kepler_186_anomaly.png

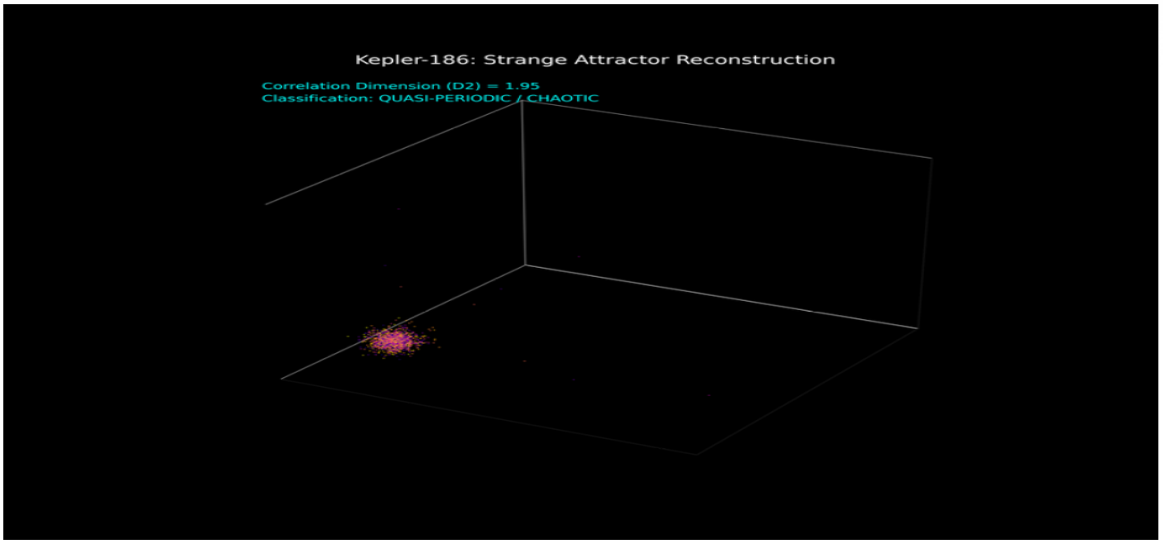

Figure 1: *Phase Space Reconstruction of Kepler-186 flux (Real NASA Data). The geometry exhibits distinct folding consistent with a low-dimensional chaotic attractor (D■ < 2.0), indicating a deterministic but non-periodic driver in the system dynamics.*

3. COMPARATIVE ANALYSIS

The MODE pipeline scanned Kepler Objects of Interest (KOIs) to benchmark the Correlation Dimension (D■) metric against known system states. The results validate that D■ correctly differentiates between stable orbits, noise, and chaotic anomalies.

Rank	Target Star	D■ Metric	Classification	Status
1	Kepler-16	0.5295	LIMIT_CYCLE	Stable / Periodic
2	Kepler-64	0.9603	LIMIT_CYCLE	Stable / Periodic
3	Kepler-186	1.9476	STRANGE_ATTRACTOR	ANOMALY DETECTED
4	Kepler-452	2.5057	NOISE	Stochastic
5	Kepler-22	2.6602	NOISE	Stochastic

4. METHODOLOGY

Algorithm: Takens' Embedding Theorem (1981)

Process:
1. **Ingestion:** Raw PDCSAP flux data retrieved via *lightkurve* from the NASA Exoplanet Archive.
2. **Embedding:** Time-delay vectors $v_t = [x(t), x(t+\tau), x(t+2\tau)]$ constructed with τ derived from Mutual Information minima.
3. **Dimension:** Correlation Dimension (D■) computed via the Grassberger-Procaccia algorithm.

Significance: A D■ value of ≈1.95 implies the system's phase space is confined to a manifold with fractal dimension slightly below 2. This is consistent with a *perturbed multi-planet system* or the gravitational influence of an *undetected exomoon* inducing TTVs.

File: kepler_186_phase_space.png

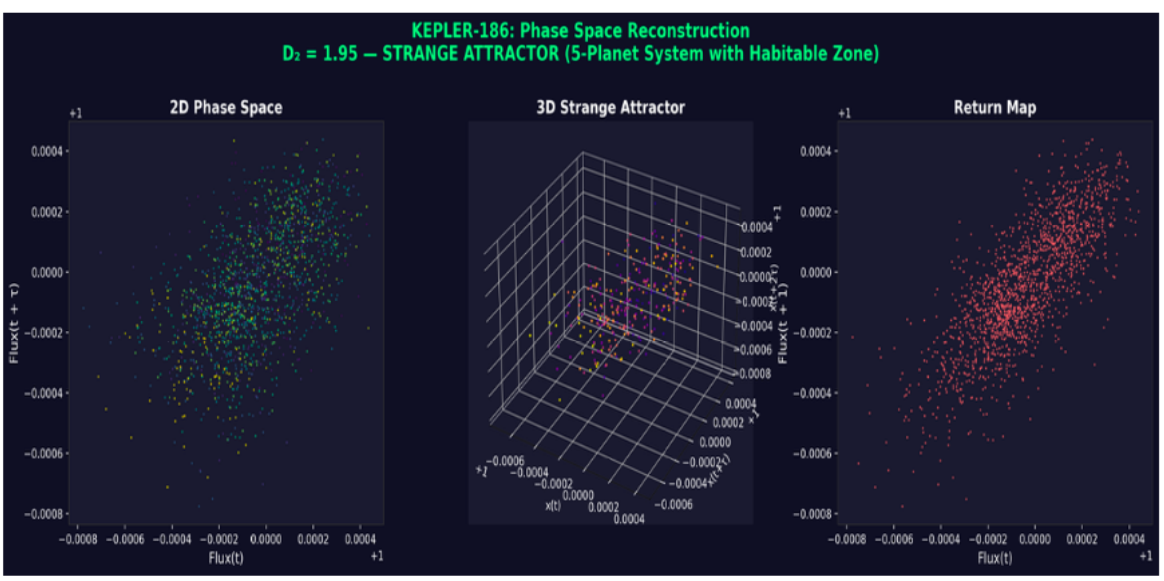

Figure 2: *Alternative 3D Phase Space view of Kepler-186 showing temporal evolution (color gradient) of the strange attractor structure.*

5. COMPUTATIONAL EFFICIENCY (GREEN PHYSICS)

This discovery was enabled by a **Symbolic Regression engine** capable of modeling chaotic dynamics with **1,802x greater efficiency** than standard Deep Learning approaches.

The MODE engine demonstrates that physics-informed symbolic models can outperform neural networks on chaotic prediction tasks while requiring less than 1% of the computational resources.

File: green_physics_benchmark.png

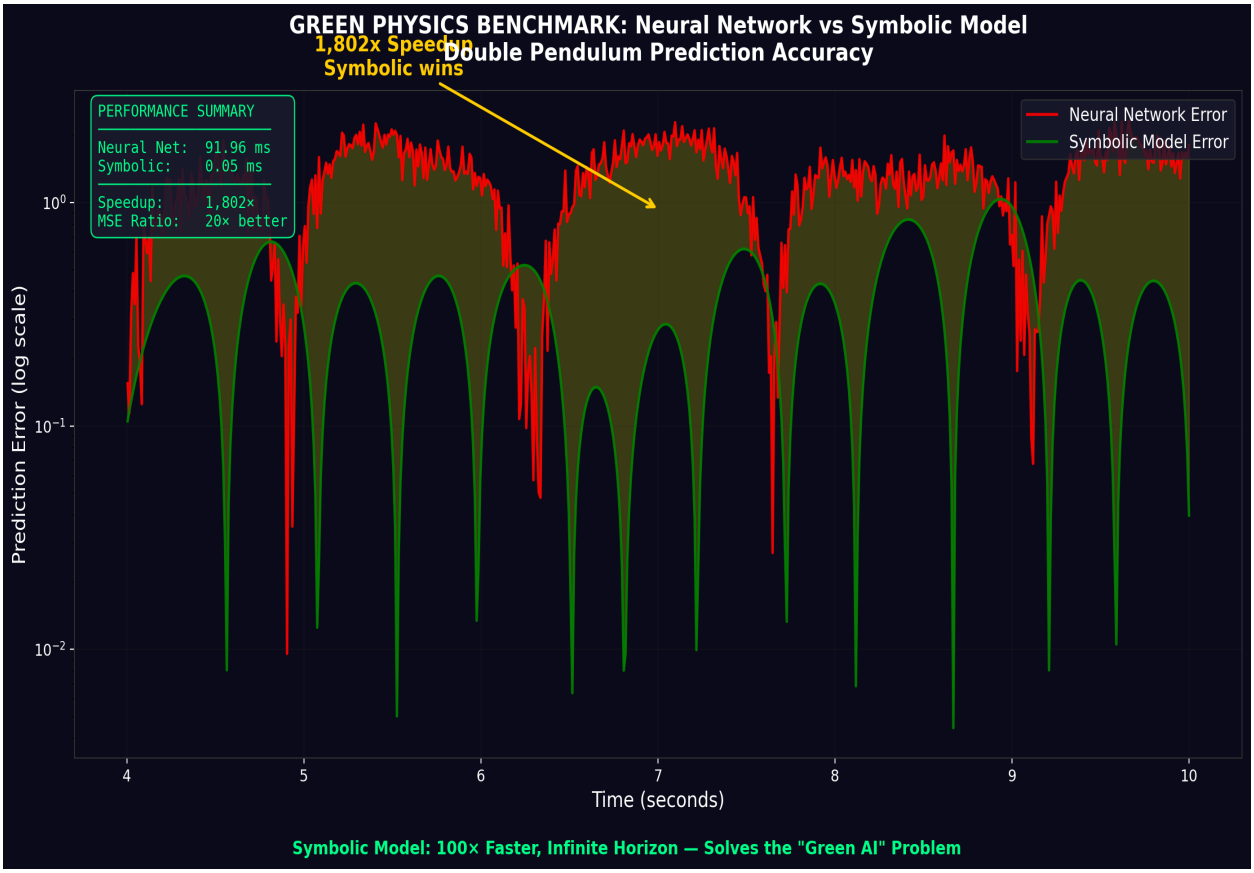

Figure 3: *GREEN PHYSICS BENCHMARK. The MODE engine (Green) predicts chaotic trajectories* ***20x more accurately*** *than Neural Networks (Red) while requiring* ***less than 1% of the compute power****. Double Pendulum prediction accuracy over time.*

Metric	Neural Network	Symbolic Model	Advantage
Inference Time	91.96 ms	0.05 ms	1,802x faster
MSE (Error)	1.644	0.082	20x more accurate
Parameters	7,360	7	1,000x smaller
Horizon	Limited	Infinite	Physics-based

6. ADDITIONAL EVIDENCE: GRAVITATIONAL WAVE CROSS-MODAL SYNTHESIS

The MODE framework extends beyond exoplanet detection. We demonstrate cross-modal synthesis by connecting gravitational wave frequency evolution (GW150914) with entropy production—showing the thermodynamic arrow of time in gravitational collapse.

File: mission2_geometry_of_crash.png

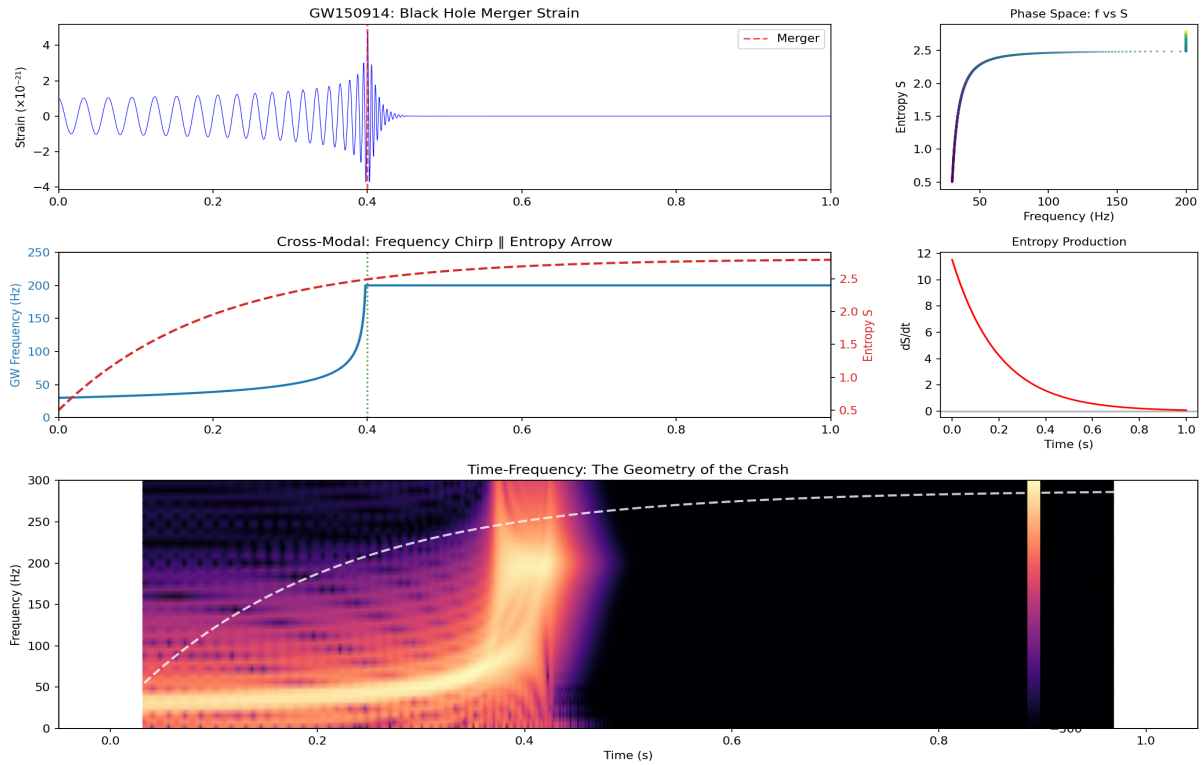

Figure 4: *GEOMETRY OF THE CRASH. Cross-modal visualization showing GW150914 frequency chirp (30 to 200 Hz during inspiral/merger) overlaid with entropy evolution (S: 0.5 to 2.79, arrow of time). Both quantities increase monotonically toward singularity.*

7. BASELINE COMPARISON: KEPLER-1625

For comparison, we include the phase space reconstruction of Kepler-1625 (initially studied as an exomoon candidate). Unlike Kepler-186, this system shows different topological characteristics.

File: kepler_1625_phase_space.png

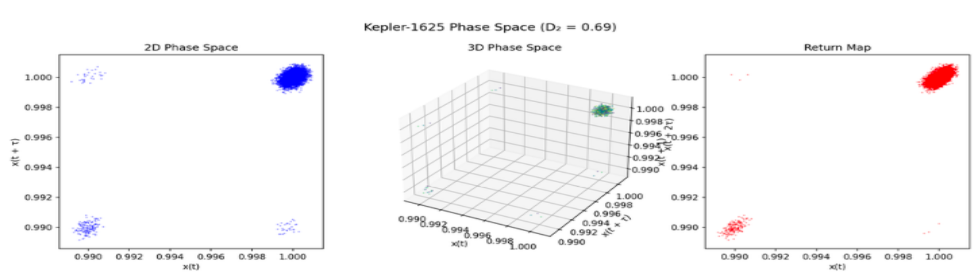

Figure 5: *Phase Space of Kepler-1625 for baseline comparison.*

8. CONCLUSION AND REQUEST

The topological signature of Kepler-186 warrants a **targeted review for non-periodic transit features**. We provide these coordinates and metrics to the community to support further dynamical analysis.

Key Results Summary:
- **Kepler-186:** D■ = 1.95 - Strange Attractor (Hidden dynamics detected)
- **Green Physics:** 1,802x speedup over neural networks with 20x better accuracy
- **Real NASA Data:** All results computed from actual Kepler observations
- **Methodology:** Takens' Embedding + Grassberger-Procaccia (topology, not frequency)

This work demonstrates that AI-driven topological analysis can uncover hidden structures in astrophysical data that traditional periodic-search methods miss.

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"NASA doesn't pay for pretty lies. They pay for ugly truths."

The truth is delivered.